Radian measure

1

a $\frac{\pi}{180}$

b $\frac{180}{\pi}$

2

a 2π radians

b π radians

c $\frac{\pi}{2}$ radians

d $\frac{5\pi}{4}$ radians

e $-\frac{5\pi}{3}$ radians

f $\frac{28\pi}{45}$ radians

g $\frac{29\pi}{180}$ radians

h 2.81 radians

3

a 60°

b 30°

c 45°

d 270°

e 225°

f -120°

g -300°

h 240.6°

4

a True

b False. 60° must equal to $\frac{\pi}{3}$ radians, because it is $\frac{1}{3}$ of 180°.

c False. 30° must equal to $\frac{\pi}{6}$ radians, because it is half of 60°.

d True

e True

f False. 225° must be equal to $\frac{5\pi}{4}$ radians, because it is five groups of 45°.

Coterminal angles

5

a $\frac{21\pi}{8}$

b $-\frac{11\pi}{8}$

6

a $\frac{25\pi}{8}$

b $-\frac{7\pi}{8}$

7

a $\frac{4\pi}{3}$

b $-\frac{8\pi}{3}$

8
a Quadrant 3 b $\frac{\pi}{3}$ c $\frac{\pi}{3}$

9
a Quadrant 3 b $\frac{\pi}{7}$ c $\frac{\pi}{7}$

10
a 454° b -266° c Quadrant 2

11
a 302° b -418° c Quadrant 4

12
a 129° b 29°

13
a $\frac{\pi}{4}$ b $\frac{\pi}{3}$

14
a $360n$ b $360n$

15
a $2\pi n$ b $2\pi n$

Reference angles

16
a 69° b 17° c 85°

17
a
i 193° ii 13°
b
i 159° ii 21°

18 30°

20 30°

21

a $\frac{14\pi}{13}$

b $-\frac{12\pi}{13}$

22

a $\frac{6\pi}{7}$

b $-\frac{8\pi}{7}$

23

a $\frac{21\pi}{11}$

b $-\frac{\pi}{11}$

24 $\frac{\pi}{11}$

25 $\frac{\pi}{7}$

26 $\frac{\pi}{5}$

27 $\frac{\pi}{3}$ radians

28

a $\frac{10\pi}{9}$ radians

b $\frac{\pi}{9}$ radians

29

a $\theta = \frac{\pi}{17}, \frac{16\pi}{17}$

b $\theta = \frac{18\pi}{17}, \frac{33\pi}{17}$

c $\theta = \frac{\pi}{17}, \frac{33\pi}{17}$

d $\theta = \frac{18\pi}{17}, \frac{16\pi}{17}$

e $\theta = \frac{\pi}{17}, \frac{18\pi}{17}$

f $\theta = \frac{16\pi}{17}, \frac{33\pi}{17}$

30

a $\frac{4\pi}{13}$

b

i $\alpha = \frac{9\pi}{13}, \frac{4\pi}{13}$

ii $\alpha = \frac{9\pi}{13}, \frac{17\pi}{13}$

iii $\alpha = \frac{9\pi}{13}, \frac{22\pi}{13}$

31

a $\frac{\pi}{17}$

b

i $\alpha = \frac{18\pi}{17}, \frac{33\pi}{17}$

ii $\alpha = \frac{18\pi}{17}, \frac{16\pi}{17}$

iii $\alpha = \frac{18\pi}{17}, \frac{\pi}{17}$

32

a $\frac{2\pi}{17}$

b

i $\alpha = \frac{32\pi}{17}, \frac{19\pi}{17}$

ii $\alpha = \frac{32\pi}{17}, \frac{2\pi}{17}$

iii $\alpha = \frac{32\pi}{17}, \frac{15\pi}{17}$

33

a $\frac{8\pi}{17}$

b

i $\theta = \frac{25\pi}{17}, \frac{26\pi}{17}$

ii $\theta = \frac{8\pi}{17}, \frac{26\pi}{17}$

iii $\theta = \frac{8\pi}{17}, \frac{25\pi}{17}$

34

a $\frac{2\pi}{17}$

b

i $\theta = \frac{2\pi}{17}, \frac{15\pi}{17}$

ii $\theta = \frac{2\pi}{17}, \frac{32\pi}{17}$

iii $\theta = \frac{15\pi}{17}, \frac{32\pi}{17}$

35

a $\frac{5\pi}{13}$

b

i $\theta = \frac{5\pi}{13}, \frac{8\pi}{13}$

ii $\theta = \frac{8\pi}{13}, \frac{18\pi}{13}$

iii $\theta = \frac{5\pi}{13}, \frac{18\pi}{13}$